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Batch C2 PRN-1262252502

Data Structures Assignment 8(linear Queue)

Code:

```
#include <stdio.h>
```

```
#define MAX 5
```

```
struct Job
```

```
{  
    int id;  
    char title[50];  
};
```

```
struct Job queue[MAX];
```

```
int front = -1, rear = -1;
```

```
void enqueue()
```

```
{  
    if (rear == MAX - 1)  
    {  
        printf("Queue Overflow! Print queue is full.\n");  
        return;  
    }
```

```
    printf("Enter Job ID: ");
```

```
    scanf("%d", &queue[++rear].id);
```

```
printf("Enter Document Title: ");  
scanf(" %[^\\n]", queue[rear].title);
```

```
if (front == -1)  
    front = 0;
```

```
printf("Print job added successfully.\\n");  
}
```

```
void dequeue()  
{  
    if (front == -1 || front > rear)  
    {  
        printf("Queue Underflow! No print jobs available.\\n");  
        return;  
    }  
}
```

```
printf("Processed Job ID: %d\\n", queue[front].id);  
printf("Document: %s\\n", queue[front].title);
```

```
front++;
```

```
if (front > rear)  
    front = rear = -1;  
}
```

```
void display()  
{  
    int i;
```

```
if (front == -1)
{
    printf("No Pending Print Jobs.\n");
    return;
}

printf("\nPending Print Jobs:\n");

for (i = front; i <= rear; i++)
{
    printf("Job ID: %d, Document: %s\n",
        queue[i].id, queue[i].title);
}
}

int main()
{
    int choice;

    printf("Printer Queue Manager\n");

    while (1)
    {
        printf("\n1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
```

```
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        enqueue();
        break;

    case 2:
        dequeue();
        break;

    case 3:
        display();
        break;

    case 4:
        printf("Program terminated successfully.\n");
        return 0;

    default:
        printf("Invalid choice.\n");
}

return 0;
}
```

Output:

```
dell@dell-Latitude-5420:~$ gedit ass8.c
```

```
^C
```

```
dell@dell-Latitude-5420:~$ gcc ass8.c
```

```
dell@dell-Latitude-5420:~$ ./a.out
```

Printer Queue Manager

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 1

Enter Job ID: 101

Enter Document Title: assignment8.pdf

Print job added successfully.

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 1

Enter Job ID: 102

Enter Document Title: assignment8a.pdf

Print job added successfully.

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 3

Pending Print Jobs:

Job ID: 101, Document: assignment8.pdf

Job ID: 102, Document: assignment8a.pdf

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 2

Processed Job ID: 101

Document: assignment8.pdf

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter your choice: 4

Program terminated successfully.